



A Prototype of Martian Terrain Detecting System

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Problem Statement

This project aims at realizing real-time monitoring of Martian terrain and returning a map to the user interface to solve the problem of Mars changing terrain due to sandstorms.

Needs

- An electronic system to automatically plan the optimized path and control the rover's motion
- A communication system to realize the exchange of information between the rover and the computer
- An user interface to show the map of Martian terrain

Design Description

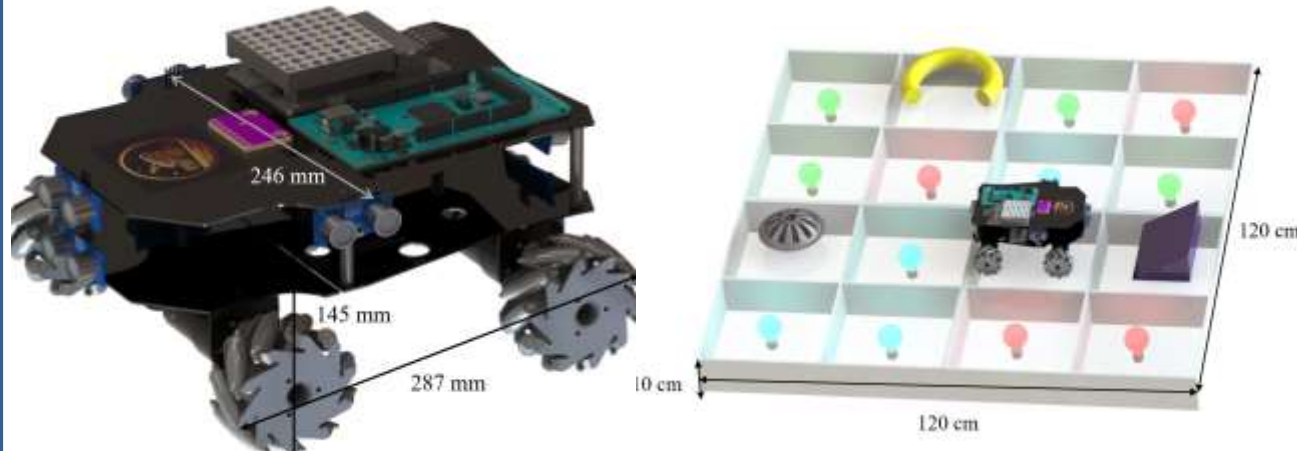


Fig. 1 The Rover and The Built Site

The prototype of the system can be separated into two parts: a rover and the user interface.

The rover is controlled by an encoder with Arduino algorithms automatically. It is also equipped with color sensors to detect the LED light on the built site, ultrasonic sensors to detect obstacles on the site, and a communication module to transmit data between the rover and the user interface. An RGB lattice is fixed on the rover's top to show the status of the terrain of the current location. The user interface is established by JavaScript programs. It consists of several desktops: a Start Screen to trigger the motion of the rover, a blank map, and real-time updated ones.

Significance of Solution

The significance lies in three parts: the plan path algorithm, the 2-dimensional map, and instant communication.

The plan path algorithm is derived from Starting Point Direction First (SPDF) algorithm, which can be well applied in closed and complex environments, making the path short.

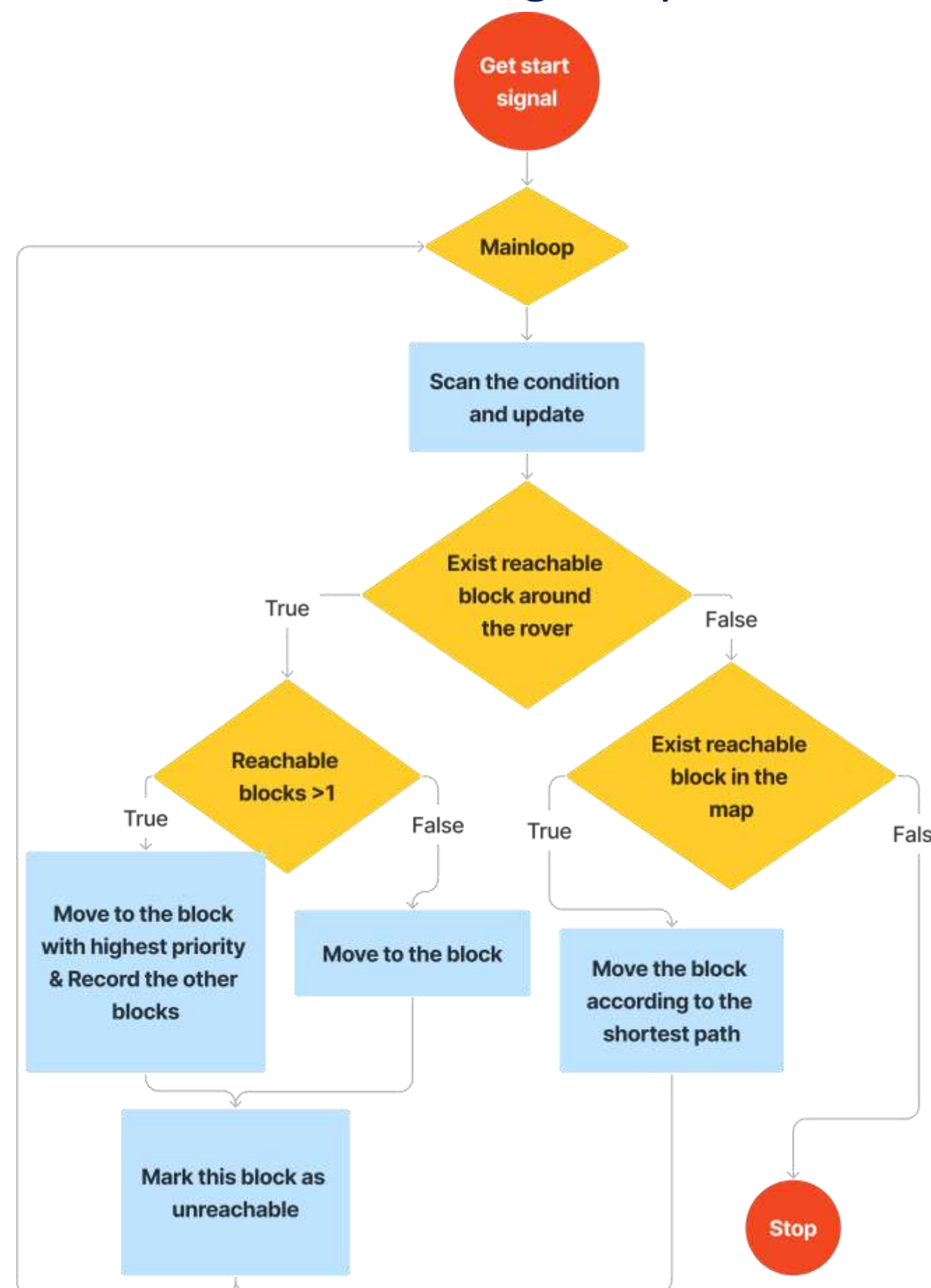


Fig. 2 Plan Path Algorithm

The generated map is composed of two dimensions: color and shape. They are measured by color ultrasonic sensors to indicate 2-dimensional data of Martian terrain: landforms beneath the rover and obstacles for it.



Fig. 3 Two-Dimensional Map

The communication is established by adopting the UDP protocol and transparent transmission mode, ensuring the high efficiency of data transmission between two parts.

Validation

For the system, the parameters of the sensors, the communication, and the algorithm are tested, including the errors in the distance measured by ultrasonic sensors, the delayed time of data transmission, and the efficiency of repeated paths in 10×10 blocks for algorithms.

According to test results, most specifications can be met.

- ✓ The errors of the ultrasonic sensors' measurements are within 0.1 cm measuring at a distance of 30.0 cm to sensitively detect obstacles.
- ✓ The delayed time of data transmission is within milliseconds to realize real-time monitoring.
- ✓ The average number of repeated blocks for the SPDF algorithm is 4.9, smaller than 88.4 for DFS (Depth First Search) algorithm, to ensure the planned paths' efficiency.

Conclusion

The system can detect the terrain of Mars in the optimized path and return a 2-dimensional map instantly to help astronauts conduct further Mars exploration in unknown areas. It shows user-friendliness with a well-designed user interface and improves the exploration safety and efficiency.

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Reference

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